

Release checklist

Repository source: `RELEASE_CHECKLIST.md`

Contents

A. Mathematical claim	1
B. Exact certificate and audits	1
C. Paper	2
D. Documentation	2
E. Metadata and legal	2
F. Continuous integration and portability	2
G. Release assets	3
H. Public wording and approval	3
Release decision	3

Candidate version: `v1.0.0-rc1` **Candidate date:** unreleased **Checklist updated:** September 19, 2026 **Audited candidate commit:** `892d4bc6fad07950e78da66e45b95063a6415af6` **Hosted workflow:** <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>

A checked item means that evidence exists in the current construction checkpoint. It does not authorize publication by itself.

A. Mathematical claim

- ☒ Exact theorem numerator and denominator are fixed in the authoritative certificate.
- ☒ Lower and near-Jung circumradius branches cover the full Jung interval.
- ☒ `03B02` is the exact lower-branch bottleneck.
- ☒ Near-Jung handoff margin is strictly positive in exact arithmetic.
- ☒ Width scaling restores the factor d^3 .
- ☒ Paper states no sharpness, equality, minimizer, or Meissner-extremality claim.
- ☐ Independent subject-matter reviewer has checked the mathematical reduction.

The unchecked external-review item is not required to label a release as a non-peer-reviewed research draft, but its absence must remain prominent.

B. Exact certificate and audits

- ☒ Standalone verifier has no private-repository dependency.
- ☒ Immutable proof objects have SHA-256 fingerprints.
- ☒ Exact certificate reconstructs byte identically.
- ☒ Independent binary64 Python audit passes.
- ☒ Independent base-R audit passes on the named author's machine.
- ☒ Independent base-R audit passes in hosted R 4.6.1.
- ☒ Ten reconstruction/regression tests pass.
- ☒ Seven adversarial mutations are rejected.
- ☒ Export-equivalence record checks ten theorem-critical fields.
- ☒ Complete replay passes from a fresh clone of the staging remote at commit `892d4bc6fad07950e78da66e45b95063a6415af6`.
- ☐ Complete replay passes from the immutable final release tag on a fresh machine.

C. Paper

- ☒ Lean Paper v1 contains only the proof supporting this coefficient.
- ☒ The theorem coefficient and proof-critical constants are generated from the certificate.
- ☒ Paper/certificate reconciliation passes.
- ☒ Reference PDF builds and has been visually inspected page by page.
- ☒ Paper limitations and computer-assisted status are explicit.
- ☒ Sole author and author order approved: Bruce J. Swihart.
- ☒ No institutional affiliation is asserted; correspondence is directed to the repository issue tracker; no ORCID is asserted.
- ☒ Candidate repository URL and version inserted; final release date and any archival identifier remain intentionally unassigned.
- ☐ Final PDF approved by every named author.

D. Documentation

- ☒ Reproducibility guide.
- ☒ Verification-status page.
- ☒ Trust-boundary document.
- ☒ Claims-and-limitations document.
- ☒ Certificate specification.
- ☒ Independent-audit record.
- ☒ Mathematical overview.
- ☒ Claim-dependency graph.
- ☒ Proof ledger and internal proof audit.
- ☒ Source ledger with dated literature check.
- ☒ Exact-constants appendix generated from the certificate.
- ☒ Certificate-architecture appendix.
- ☒ Replay-transcript appendix.
- ☒ AI-assistance and provenance disclosure.
- ☒ Clean-source, hosted, and fresh-clone record.
- ☒ Draft release notes and release checklist.
- ☒ Every Markdown source has a checksummed PDF counterpart.
- ☒ Remote evidence is indexed in `certificate/evidence/rc1_remote_audit.json`.
- ☐ Final copy edit and link review by the named author.

E. Metadata and legal

- ☒ Repository name confirmed: `swihart/minvol-degree3-contact-bound`.
- ☒ Repository description confirmed.
- ☒ Split license selected and `LICENSE` added: CC BY 4.0 for paper/prose and MIT for software/build infrastructure.
- ☒ `CITATION.cff` added and checked by the metadata preflight.
- ☒ Preferred paper citation finalized for the release candidate.
- ☒ Candidate version changed to `v1.0.0-rc1`.
- ☐ `RELEASE_DATE` changed from `UNRELEASED` to the approved final date.
- ☐ Candidate version changed from `v1.0.0-rc1` to final tag `v1.0.0`.
- ☐ DOI or archival identifier added only after one exists.
- ☒ Copyright and third-party license boundary reviewed and documented.

The unchecked date, final-tag, and archival items cannot be completed before explicit final approval.

F. Continuous integration and portability

- ☒ Workflow has separate Python, base-R, and document jobs.
- ☒ Hosted document job installs Poppler and checks fonts, text, and page validity.
- ☒ Local Mac without Poppler is documented rather than treated as a theorem failure.
- ☒ PDF byte identity across platforms is not required.
- ☒ Three hosted jobs are green on audited candidate commit `892d4bc6fad07950e78da66e45b95063a6415af6`.

- ☒ The successful candidate run is recorded at <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>.
- ☒ A fresh clone of that commit completed the Python and base-R replay, rebuilt the paper and 22 documentation PDFs, and ended clean.
- ☐ Three hosted jobs are green on this evidence-bearing commit after it is pushed.
- ☐ Three hosted jobs are green on the immutable release tag.
- ☐ Clean tagged-checkout replay is recorded in `CLEAN_CLONE_CHECK.md`.

G. Release assets

- ☐ Paper PDF asset built from the tagged checkout.
- ☐ Paper-source archive built from the tagged checkout.
- ☐ Standalone certificate archive built from the tagged checkout.
- ☐ Documentation archive built from the tagged checkout.
- ☐ Replay-transcript archive built from the tagged checkout.
- ☐ One release-asset `SHA256SUMS.txt` generated and independently checked.
- ☐ Every downloaded asset reverified in a clean directory.
- ☐ Release notes PDF generated from the final Markdown source.

H. Public wording and approval

- ☒ Draft wording says “computer-assisted” and “not peer reviewed.”
- ☒ Draft wording does not say “sharp,” “optimal,” “proved Meissner,” or “independently certified.”
- ☒ Earlier public spectral repository is clearly separate.
- ☒ Private sequel results and history are absent.
- ☒ Named author approved creation and use of the private staging remote.
- ☐ Named author approves README.
- ☐ Named author approves theorem wording.
- ☐ Named author approves final PDF.
- ☐ Named author approves release notes.
- ☐ Named author approves announcement text.
- ☐ Named author explicitly approves changing repository visibility to public.
- ☐ Named author explicitly approves creation of the GitHub Release.

Release decision

Current decision:

DO NOT RELEASE YET. The mathematical, documentation, authorship, citation, licensing, hosted-candidate, and fresh-clone gates have passed for commit `892d4bcfad07950e78da66e45b95063a6415af6`. The evidence commit, final copy approval, date/version transition, immutable tag, tagged replay, release assets, visibility change, and explicit publication approvals remain open.